

WHEN TO STEER: SENSITIVITY-CALIBRATED PROBE GUIDANCE FOR CONTROLLABLE MUSIC DIFFUSION

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ABSTRACT

Probe-guided music diffusion steers pitch content with a lightweight predictor that reads the latent, but in existing methods the sampling steps at which the guidance is applied are fixed in advance. We measure, along real Stable Audio Open trajectories, how probe decodability and the effect of a guidance step change during denoising. Short guidance bursts change melodic coherence (a pitch-class match rate) most at 54% of sampling, whereas probe F_1 reaches 95% of its maximum only at 88%. Sensitivity-calibrated probe guidance (SCPG) allocates a fixed update budget to the steps with the largest measured coherence gain on a development set. With 15 updates it raises melodic coherence on held-out prompts from 0.363 for the corrected prior schedule to 0.434; CLAP does not differ significantly from the prior schedule and FAD is numerically similar. On Stable Audio 3 the same calibration selects an earlier interval and raises coherence from 0.28 for the prior schedule to 0.44, while CLAP decreases from 0.375 (unguided) to 0.335. The effective interval differs between the two models; measuring it costs one burst sweep on five development prompts.

Index Terms— controllable music generation, latent diffusion, training-free guidance, probing, guidance scheduling

1. INTRODUCTION

Text-to-music diffusion models such as Stable Audio Open [1] (SAO) produce high-quality audio but offer only coarse control through the text prompt. Trained conditioning offers melody control [2, 3]. Inference-time guidance adds it without retraining, by evaluating a differentiable target loss on the partially denoised sample and using its gradient to update the latent [4, 5, 6, 7, 8], or by optimising the sampler through its own trajectory [9]. For music transformers, probe-steered activations with a self-monitoring signal [10] are the closest relative of the reliability signal used here. For latent audio diffusion, back-propagating through the audio decoder is expensive, so two recent works read the control directly from the latent with a small network: latent-control heads (LatCH) [11] and pitch-class probes [12]. Both operate on SAO, both require only probe evaluations, and both apply guidance on a fixed interval of sampling steps.

The two methods apply guidance at different stages of denoising. LatCH restricts its “selective” guidance to the first 20% of steps, arguing that later interventions drift off the data manifold [11]; [12] waits until 40% of denoising is complete and then intervenes every second step, arguing that early latents are too noisy for the probe. Neither work reports an ablation of the interval. The question is not specific to these two methods: limited-interval guidance matters

for classifier-free guidance in image diffusion [13], and training-free guidance frameworks expose the interval as a hyper-parameter [7]. For probe-based latent guidance of music the answer is not obvious, because the probe’s information content, the effect of a gradient step and its cost in audio quality all change along the trajectory, and not necessarily together.

This paper studies that question on SAO with the pitch-class probe of [12] and turns the measurements into a schedule. Our contributions are:

1. A comparison, along real sampling trajectories, of probe decodability with the interventional effect of a short guidance burst at each position (Secs. 4.1–4.2).
2. SCPG, which allocates a fixed guidance budget to the steps of largest measured coherence gain on a development set, with reliability-based strength scaling and an online reliability threshold studied as ablations (Sec. 2).
3. An evaluation on two model configurations, SAO 1.0 and the Stable Audio 3 base model, reporting the coherence gains and their cost in prompt adherence (Secs. 4.3–4.5).

We also correct the target/latent alignment of the public implementation of [12] (Sec. 3); all numbers use the corrected code. Code, per-run results and audio examples are online.¹

2. METHOD

2.1. Probe-guided sampling

SAO [1] denoises a latent $\mathbf{z} \in \mathbb{R}^{64 \times T}$ ($T=1024$ frames at 21.5 Hz) with a diffusion transformer and an EDM-style sampler [14]; we use $N=50$ steps and classifier-free guidance [15] of 4.0. Step t ($t=0, \dots, N-1$) refers to the latent $\mathbf{z}_t$ returned by the $(t+1)$ -th scheduler update, to which guidance is applied before the next denoising step; positions are reported as denoising progress $s=(t+1)/N$, the fraction of updates completed. A convolutional probe f_ϕ with 125k parameters, trained on VAE latents of MAE-STRO [16] with Gaussian-noise augmentation, maps a latent to frame-level pitch-class probabilities $p_t=f_\phi(\mathbf{z}_t) \in [0, 1]^{12 \times T}$ [12]. Given a target pitch-class roll $\mathbf{y}^* \in \{0, 1\}^{12 \times T}$, the guidance loss is the binary cross-entropy $\mathcal{L}_t=\text{BCE}(p_t, \mathbf{y}^*)$ restricted to the audible frames, with gradient $\mathbf{g}_t=\nabla_{\mathbf{z}_t} \mathcal{L}_t$, and the latent is updated with the RMS-normalised, clamped step of [12]:

$$\mathbf{z}_t \leftarrow \mathbf{z}_t - \text{clamp}\left(\lambda_t \frac{\text{rms}(\mathbf{z}_t)}{\text{rms}(\mathbf{g}_t)} \mathbf{g}_t, \pm \alpha \text{rms}(\mathbf{z}_t)\right), \quad (1)$$

with $\alpha=0.05$. In [12], $\lambda_t=\lambda$ is constant and the update fires on a fixed set of steps $\mathcal{T}$ (every second step after 40% of denoising,

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¹<https://stoneyey.github.io/mic-pitch-class-steering/>

i.e. $K=15$ updates); LatCH guides during the first 20% of steps. Everything that follows changes only $\mathcal{T}$ and λ_t .

2.2. Sensitivity calibration and SCPG

SCPG chooses $\mathcal{T}$ from a measurement. On development prompts, each trial is generated twice from the same seed, once unguided and once with guidance at three consecutive steps $\{p, p+1, p+2\}$ only (a *burst*, $\lambda=0.10$), and the change in melodic coherence of the final audio, $\Delta C(p)$, is averaged over 50 trials. Bursts are placed at 12 positions $p \in \{2, 6, \dots, 46\}$; the measured values are linearly interpolated onto all N steps and smoothed with a three-step moving average, giving a per-step steering-sensitivity profile $w(t)=\Delta C(t)$. $\mathcal{T}$ is the set of K steps with the largest $w(t)$ (ties towards the earlier step). Ranking steps by single-burst effects is a heuristic; the full-budget evaluation of Sec. 4.3 tests whether it yields an effective multi-step schedule. For comparison we also calibrate on decodability, $w(t)=F_1(t)$, the probe’s micro- F_1 against pYIN labels along real trajectories (Sec. 4.1), which tests directly whether guidance should be placed where the probe is most accurate or where its gradient changes the output most. Calibration costs one burst sweep per model (12 positions $\times$ 50 development trials plus references: 650 generations, 0.8 GPU-hours on an RTX 5080 for SAO, 0.5 for Stable Audio 3). Once calibrated, SCPG requires no online schedule search and uses the same number of guidance updates as the fixed-schedule baselines.

2.3. Reliability-based variants

As a target-free reliability proxy, available at every step, we use the concentration of the probe’s output. The 12 sigmoid outputs are independent and do not form a distribution over classes, and a probe that confidently predicts no note anywhere on pure noise would look certain; we therefore normalise the outputs of each frame, $q_{t,f}=p_{t,f}/\sum_c p_{t,f,c}$, and measure how peaked they are:

$$R_t = 1 - \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \frac{H(q_{t,f})}{\log 12} \in [0, 1], \quad (2)$$

where $\mathcal{F}$ is the set of audible frames. R_t is insensitive to the overall magnitude of p_t and costs one probe forward pass (<1 ms). Two variants use it. *R-scaling* sets $\lambda_t=\lambda R_t/\bar{R}$, where $\bar{R}$ is the mean reliability over the guided steps on the development set; this matches the mean guidance strength on the development set, while the realised mean may differ on held-out samples. The *online R-threshold* guides at step t iff $R_t \geq \eta$ and fewer than K updates have been used, with η chosen on the development set so that the mean number of updates is K ; it needs no sensitivity profile and is combined with *R-scaling*. All variants keep Eq. (1) unchanged.

3. EXPERIMENTAL SETUP

Base model and probe. SAO 1.0 in fp16, 50 sampling steps, CFG 4.0, 5-second outputs; the CNN probe checkpoint of [12] (micro- F_1 0.66 on held-out MAESTRO clips) is used unchanged. The main study uses SAO 1.0 because both schedules under study and the probe were developed on it, so that only the schedule differs between methods; Sec. 4.5 repeats the analysis on Stable Audio 3. The SAO pipeline always denoises a 1024-frame latent (47.6 s) and crops the decoded audio to the requested length. The public implementation of [12] constructed the target over the full latent rather than the audible 5-s region, so a 4-second melody was stretched over 38 s of

latent time and the audible seconds were steered towards its first note only. All methods here place the target at the true latent frame rate of 21.5 frames/s and evaluate the loss on the 108 audible frames; *MLSP-fixed* denotes the schedule of [12] with this corrected alignment (with the original alignment it scores 0.259 against the melody and 0.68 against its first pitch class alone).

Prompts, melodies, seeds. All profiles, schedules and thresholds ($w(t)$, $\mathcal{T}$, η , $\bar{R}$) are calibrated exclusively on five development prompts; the evaluation prompts are held out and never used for schedule selection. The test set has 15 piano-centred prompts (the 9 of [12] plus 6 new), 5 melodies of 7–8 quarter notes at 120 BPM (3.5–4 s of each 5-s clip: the ascending, descending and alternating patterns of [12], a repeated-note pedal and a leap-heavy zigzag) and 3 seeds, i.e. 225 generations per method, paired across methods.

Metrics. *Melodic coherence* is the pitch-class match rate of pYIN [17] frames inside target-active regions, as in [12]; it measures pitch-class agreement, not melodic perception as a whole. *Chroma cosine* is the mean cosine between the CQT chroma of the output and the binary target; *CLAP* is the text–audio cosine of LAION-CLAP (clap-htsat-unfused) [18]; *FAD* is the Fréchet distance [19] between CLAP embeddings of each method’s outputs and of 600 MAESTRO clips. We report means with bootstrap 95% CIs and paired Wilcoxon signed-rank tests.

Baselines. All guided methods use $K=15$ updates and $\lambda=0.05$. *Early* (steps 0–14) is a budget-matched early schedule inspired by LatCH: LatCH’s own window is the first 20% of steps (10 of 50), so *Early* covers the first 30% to match the budget, and it tests early placement under our update rule. The other schedules are *Mid* (18–32), *Late* (35–49), *Uniform* (every ~ 3.5 steps), *MLSP-fixed* (20, 22, $\dots$, 48), SCPG (top- K of the ΔC profile, constant λ) and its *R-scaled* variant, the decodability-calibrated ablation (top- K of F_1 with *R-scaling*) and the online *R-threshold* (with *R-scaling*); unguided SAO is the reference. Paired Wilcoxon tests are Holm-corrected within each family of nine comparisons against a reference method and metric.

4. RESULTS

4.1. Decodability along real trajectories

Fig. 1(a) plots, over 50 unguided development trajectories, the probe’s micro- F_1 against pitch-class labels transcribed from the generated audio by pYIN (the estimator behind the coherence metric), together with R_t of Eq. (2). Decodability is near zero in the early latents and rises essentially monotonically, reaching 95% of its maximum only at 88% of the trajectory. R_t , computed without any target, follows the same curve (Pearson 0.96 across steps; median 0.85 within a trajectory). Part of this agreement is the shared trend: at a fixed step, across samples, R_t predicts a sample’s F_1 only weakly early and moderately late (median Spearman 0.27; 0.57 over the last 15 steps), which bounds what an online rule can read from it. The Gaussian-noise proxy used to train the probe [12] reproduces the curve when the noise level is matched to each step.

4.2. Steering sensitivity

Fig. 1(b) shows the interventional profile of Sec. 2.2: the change in coherence from a three-step burst starting at one position, paired against the same-seed unguided generation. The gain peaks at $s=0.54$ of sampling and is several times smaller both early and late; a burst inside the first 20% of steps, the interval of selective guidance in [11], changes coherence by less than a tenth of the mid-sampling

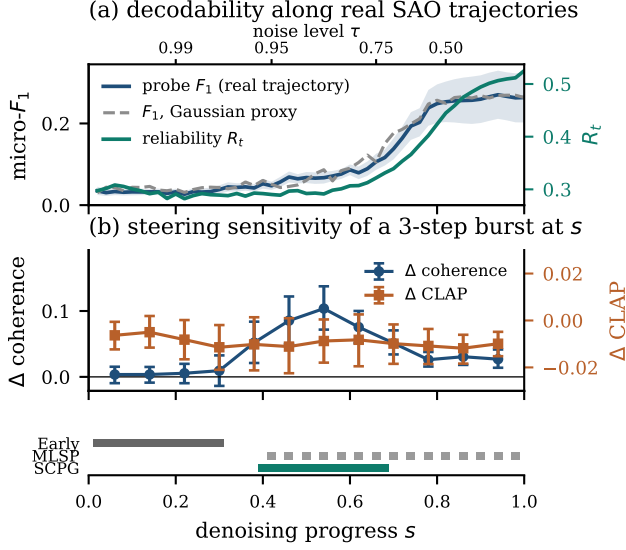


Fig. 1. (a) Probe decodability along 50 real SAO development trajectories (5 prompts $\times$ 10 seeds): micro- F_1 against pYIN labels of the generated audio, the target-free reliability R_t of Eq. (2), and the Gaussian-noise proxy of [12] at matched noise level. (b) Steering sensitivity: change in coherence and in CLAP from a burst of three updates at steps $p, p+1, p+2$ (mean, 95% CI, 50 paired development trials), plotted at the progress of its first update. Top axis: noise level of the latent as $\tau = \sigma / (1 + \sigma)$, σ the noise-to-signal rms ratio. Bottom: guided steps of the Early, MLSP-fixed and SCPG schedules.

effect. This ordering is the reverse of decodability, which is highest at the end of sampling. A burst at $s=0.94$ lowers the probe loss on the final latent more than one at $s=0.54$ (BCE change -0.14 vs -0.11) while changing the audio less (log-mel distance to the unguided output 0.21 vs 0.53). A single burst lowers CLAP slightly at every position (mean Δ CLAP between -0.012 and -0.005 relative to the unguided output; Fig. 1(b), right axis); Sec. 4.3 reports CLAP and FAD for the full guidance budget.

The peak is stable. Over 5000 bootstrap resamples of the 50 trials the argmax lies at $s=0.54$ in 92% and within 0.46 – 0.62 in all of them; the five leave-one-prompt-out profiles peak between $s=0.54$ and 0.62 ; and the mid positions beat the early ones in 88% of individual trials (paired Wilcoxon $p < 10^{-7}$). A control burst in a random direction with the same update norm, applied at the peak position, changes coherence by 0.003 (n.s.) against 0.104 for the probe gradient (paired $p < 10^{-6}$), so the effect is directed. Repeating the sweep at the $\lambda=0.05$ of the main experiments gives the same profile (peak at $s=0.54$, $\Delta C=0.099$ vs 0.104 , Pearson 0.99 between the curves; the top-15 steps shift by one, to 18 – 32). The timing conclusion holds for the tested update rule and step size: with $\lambda=\alpha=0.2, 0.5$ and 1.0 the early window’s development-set coherence rises from 0.12 (unguided) to $0.23, 0.29$ and 0.41 while CLAP falls from 0.207 to $0.187, 0.184$ and 0.173 , so early placement can steer at larger step sizes at a cost in prompt adherence; under the common update rule and nominal guidance strength the measured coherence gain is largest in the mid window.

Table 1. Test-set means, all guided methods with $K=15$ updates ($n=225$ per row; MLSP-fixed has 224 valid coherence values and 225 CLAP values, and paired tests use complete pairs); schedules calibrated on the disjoint development set. Bootstrap 95% CI half-widths ≤ 0.035 (coherence) and ≤ 0.017 (CLAP). $+/-$: coherence significantly higher/lower than MLSP-fixed (paired Wilcoxon, Holm-corrected, $p < 0.05$). MLSP-fixed guides every second step from 20 to 48. FAD: CLAP-embedding Fréchet distance to real piano (upward-biased at this sample size); #u.: probe updates.

Method	Coh. $\uparrow$	Chroma $\uparrow$	CLAP $\uparrow$	FAD $\downarrow$	#u.
SAO (unguided)	0.126 ⁻	0.22	0.307	0.82	0
Early (0–14)	0.166 ⁻	0.28	0.312	0.81	15
Mid (18–32)	0.424 ⁺	0.54	0.317	0.77	15
Late (35–49)	0.315 ⁻	0.46	0.311	0.81	15
Uniform	0.283 ⁻	0.41	0.314	0.79	15
MLSP-fixed (20:2:48)	0.363	0.48	0.316	0.79	15
Top- $K(F_1) + R$ -scaling	0.310 ⁻	0.46	0.310	0.81	15
R -threshold (online)	0.351	0.45	0.314	0.77	15.0
SCPG: Top- $K(\Delta C)$	0.434 ⁺	0.55	0.321	0.76	15
SCPG + R -scaling	0.445 ⁺	0.56	0.324	0.75	15

4.3. Guidance window ablation and SCPG

Table 1 compares the schedules at the same budget. Placement is the dominant factor: at identical budget the early window improves little over unguided SAO (0.17 vs 0.13), the mid window reaches 0.42 , and the late and MLSP windows fall in between. SCPG selects steps 19 – 33 (18 – 32 at $\lambda=0.05$), i.e. it recovers the mid window without a window sweep, and reaches 0.434 against 0.363 for MLSP-fixed ($p < 10^{-6}$) and 0.424 for Mid ($+0.010$, $p = 0.002$). The decodability-calibrated ablation selects exactly the late window (35 – 49), because $F_1(t)$ increases monotonically; in these experiments, ranking steps by decodability favours the less effective late updates. The advantage over MLSP-fixed holds for four of the five melodies and is within noise on the alternating pattern (0.317 vs 0.346).

CLAP does not differ significantly from MLSP-fixed for any guided method (0.321 – 0.324 for the SCPG variants vs 0.316 , and 0.307 unguided), and the Fréchet distance to real piano is numerically similar (0.75 – 0.76 vs 0.79 ; from 225 samples it is upward-biased and given for orientation only). Neither metric measures perceived quality. The coherence gain is not explained by reduced voicing (pYIN-voiced fraction 0.88 under SCPG, 0.86 MLSP-fixed, 0.84 unguided), and it exceeds what the same audio would score if collapsed onto the melody’s first pitch class (0.434 vs 0.249), so the guidance follows the time-varying melody. pYIN is monophonic, so on polyphonic output the metric scores each frame’s dominant pitch; chroma agreement is reported alongside it.

R -scaling adds a small and consistent gain (0.445 vs 0.434 ; $+0.011$, better in 122 of 184 non-tied trials, $p < 10^{-5}$), but its realised mean step on the test set is 0.057 rather than 0.05 , so the gain cannot be separated from a 15% larger average step; about 86% of the improvement over MLSP-fixed comes from placement alone. On the late window R -scaling slightly reduces mean coherence, from 0.315 to 0.310 (unadjusted paired $p < 10^{-6}$). The online R -threshold, which also uses R -scaling, does not improve over MLSP-fixed (0.351 vs 0.363 , $p=0.53$); its first update falls at steps 2 – 12 and its last at step 40 on the median trial, so its budget is spread over most of the trajectory rather than concentrated in the mid window.

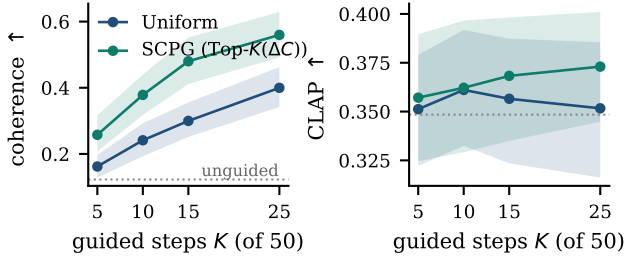


Fig. 2. Budget sweep on a paired 50-trial subset: coherence (left) and CLAP (right) against the number of guided steps, uniformly spread vs placed at the steps of highest steering sensitivity (mean, bootstrap 95% CI). Dotted: unguided.

4.4. Budget sweep

Fig. 2 varies the number of guided steps, $K \in \{5, 10, 15, 25\}$, on a paired subset of the test set (10 prompts $\times$ 5 melodies, one seed) for the uniform schedule and for SCPG with constant λ , so that only the placement of the budget differs. The sensitivity-placed budget is better at every K (paired Wilcoxon $p < 10^{-5}$): ten sensitivity-selected updates recover 95% of the coherence of twenty-five uniformly spread ones (0.38 vs 0.40), and at $K=25$ coherence reaches 0.56 against 0.40 without saturating, so the useful interval is wider than 15 steps. CLAP ranges from 0.351 to 0.373 across budgets and schedules (0.348 unguided on this subset), with overlapping CIs at every K . The sweep is single-seed and an efficiency analysis, not the primary comparison; its CLAP level differs from Table 1 because the subset’s prompts score higher for every method.

4.5. Evaluation on Stable Audio 3

We repeat both analyses on the base checkpoint of Stable Audio 3 medium [20]: a 1.4B-parameter flow-matching transformer (2.3B with autoencoder and text encoder) over 256-d latents at 10.8 frames/s, run for 50 Euler steps on its log-SNR-shifted schedule with CFG 7. The released checkpoints are post-trained with ARC [21] for 8-step sampling, which leaves only eight positions for a schedule, so we use the base model. A probe of the same architecture, trained on its latents with the recipe of [12], reaches held-out micro- F_1 0.59 (0.66 on SAO); the burst sweep and all schedules are recalibrated on five development prompts.

Fig. 3 shows the same ordering with the effective interval elsewhere. Decodability reaches 95% of its maximum only at 76% of denoising, tracked by R_t (Pearson 0.93), whereas a burst is most effective early: the gain is flat within its CI at $\Delta C \approx 0.08$ over the first 40% of steps, where the shifted schedule keeps the noise level at $\tau \geq 0.95$ and probe F_1 remains low, and falls to 0.017 in the last 20%, where the probe is most accurate. A random-direction control at the same norm gives -0.026 against 0.089 for the probe gradient (paired $p < 10^{-4}$). In noise-level terms the two intervals are adjacent rather than opposite: SAO’s mid window (steps 18–32) spans $\sigma \approx 27 \rightarrow 3.2$ ($\tau \approx 0.96 \rightarrow 0.76$), while SA3’s plateau lies at $\tau \geq 0.95$; the near-pure-noise steps ($\tau > 0.98$) are inert on SAO and effective on SA3. One possible explanation is that target-aligned perturbations introduced at high noise influence the subsequent denoising; our experiments do not identify this mechanism.

Calibration selects steps 4–18; the budget-matched early schedule (0–14) also lies inside the plateau. On 50 paired test trials (10

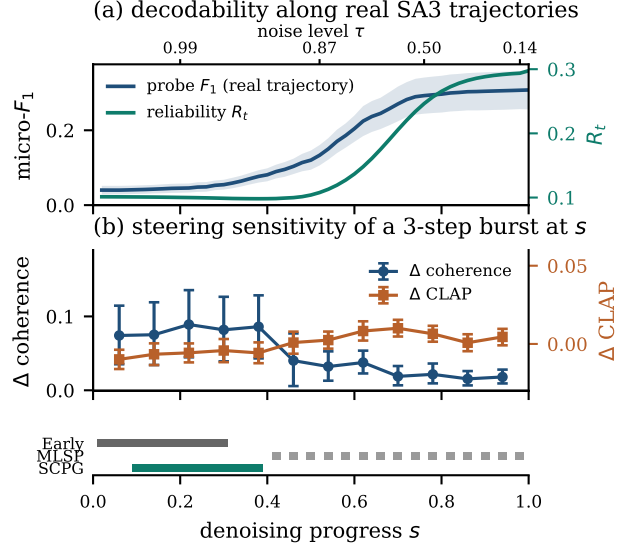


Fig. 3. The same analysis on Stable Audio 3 medium (base, 50 steps): (a) probe micro- F_1 and reliability R_t along 50 real development trajectories; (b) change in coherence and in CLAP from a three-step burst starting at one position (mean, 95% CI, 50 paired development trials). Top axis: noise level τ at each position. Bottom: guided steps of the Early, MLSP-fixed and SCPG schedules on SA3.

held-out prompts $\times$ 5 melodies, one seed; 49 valid coherence values for Mid and Late) at $K=15$, SCPG reaches coherence 0.44 (unguided 0.09) against 0.28 for the schedule of [12] (paired Wilcoxon, Holm-adjusted $p=0.002$) and 0.18 for the late window; Early, which improves little on SAO at this step size, reaches 0.44 here, within the CI of SCPG. CLAP decreases from 0.375 unguided to 0.335 under SCPG (Holm $p < 0.01$), and every guided schedule except that of [12] (0.361, n.s.) shows a similar decrease; FAD was not evaluated on SA3. The best-performing interval differs between the two evaluated configurations, and the burst sweep identifies it in both.

5. DISCUSSION AND CONCLUSION

Across the two evaluated models, the guidance positions with the largest coherence gains occur before probe decodability saturates. Image diffusion shows a related pattern: coarse content is decided at high noise [22] and intermediate latents are the most editable [23]. A development-set burst sweep identifies effective positions for allocating a fixed guidance budget. On SAO, this allocation improves coherence over the corrected prior schedule with similar CLAP and FAD; on SA3, the coherence gain is accompanied by lower CLAP. Reliability-based scaling provides a smaller additional gain whose contribution is confounded by a higher mean guidance strength on the test set, and the online reliability threshold, which follows decodability, does not improve on the prior schedule. Our conclusions are limited to pitch-class guidance applied to the latent under the tested 50-step sampling configurations, and to the automatic proxies CLAP and FAD; perceived quality remains to be assessed with listening tests.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This study used only publicly available pretrained models (Stable Audio Open 1.0 and the Stable Audio 3 medium base checkpoint, both under the Stability AI Community License) and datasets (MAESTRO), and generated synthetic audio; it involved no human or animal subjects, and no ethical approval was required.